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# **AI and Digital Dentistry in Prosthodontics: A Systematic Review**

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## **Author's contribution**

*The sole author designed, analysed, interpreted and prepared the manuscript.*

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## **ABSTRACT**

Artificial intelligence (AI) and digital dentistry have been used to bring about a transformation in the field of prosthodontics. This transformation has resulted in improvements in patient outcomes, manufacturing, diagnostics, and treatment planning. This systematic review investigates the developments, applications, challenges, and potential future trends associated with artificial intelligence-driven tools in prosthodontic practice. These tools include computer-aided design and computer-aided manufacturing (CAD/CAM) technologies, virtual articulation, 3D printing, and machine learning algorithms for individualized prosthesis design. There was a search conducted in electronic databases such as PubMed, Scopus, and Web of Science for articles that were published between the years 2015 and 2024. Only 54 of the 3,214 articles that were reviewed met the requirements to be included. The field of prosthodontics has seen significant improvements in terms of accuracy, efficiency, and patient satisfaction thanks to the implementation of digital tools and artificial intelligence. However, there are still issues regarding standardization, data privacy, and clinical validation. For the purpose of ensuring responsible and substantial application, there is a need for increased collaboration across multiple disciplines.

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**Keywords:** Artificial intelligence; digital dentistry; prosthodontics; CAD/CAM; 3D printing; machine learning.

## 1. INTRODUCTION

The area of prosthodontics stands out as a specialist within the field of dentistry because it is known for its unique ideas that frequently push the limits of imagination in order to generate results that are both practical and cosmetic for comprehensive oral rehabilitation ("Çiçek Ö, Abdulkadir A, Lienkamp S, Brox T, Olaf Ronneberger O. 3D U-Net: learning dense volumetric segmentation from sparse annotation. *Med Image Comput Comput Interv.* 2016;424–32;"). "Prosthodontics is the dental specialty that pertains to the diagnosis, treatment planning, rehabilitation, and maintenance of the oral function, comfort, appearance, and health of patients with clinical conditions associated with missing or deficient teeth and/or maxillofacial tissues and/or biocompatible substitutes," states the most recent edition of the Glossary of Prosthodontic Terms (GPT). This definition is based on the tenth edition of the GPT ("Zhang L, Zhao Y, Meng D, Cui Z, Gao C, Gao X et al. TSGCNet: Discriminative geometric feature learning with two-stream GraphConvolutional network for 3D dental model segmentation. *ArXiv:* 2012. 1369v1. Posted 26 Dec 2020. <http://arxiv.org/abs/2012.13697>. Accessed 24 June 2024;"). This expertise also include the management of maxillofacial patients, bruxism, and concerns related to the temporomandibular joint (TMJ). As a result, it offers a prosthetic solution for mouth problems that are either congenital or acquired, such as a cleft palate, oral cancer, or traumatic injuries ("Thurzo A, Jančovičová V, Hain M, Thurzo M, Novák B, Kosnáčová H et al. Human remains identification using Micro-CT, chemometric and AI methods in forensic experimental reconstruction of dental patterns after concentrated sulphuric acid significant impact. *Molecules.* 2022;27(13);").

The restoration and replacement of teeth and other oral tissues through the utilization of dental prosthesis represent the fundamental objective of this practical profession that has its origins in ancient times. Recent developments in the field of dentistry include revolutionary 3D-printed prostheses, digital dentures, and all-ceramic inlay-retained fixed dental prostheses. These advancements are in addition to the most common dental prostheses, which include dentures, crowns, bridges, dental implants, inlays, onlays, overlays, veneers, overdentures,

and implant-supported prostheses. Tools like computer-aided design and computer-aided manufacturing (CAD/CAM) systems, 3D printing, and intraoral scanners are examples of the ways in which digital technology innovations are altering the field of prosthetic dentistry ("Simonyan K, Vedaldi A, Zisserman A. Deep inside convolutional networks: visualising image classification models and saliency maps. *ArXiv:* 1312. 6034v2 [Prinprint]. Posted 19 Apr 2014. <http://arxiv.org/abs/1312.6034>. Accessed 24 June, 2024;"). These innovations utilize cutting-edge breakthroughs to improve precision, personalization, and patient results. Clinicians are now able to apply entire digital workflows in prosthodontics as a result of these innovations, which have revolutionized the design and production of dental prostheses. Their benefits include increased precision and efficiency, as well as improved patient results, which make it possible to create restorations that are highly tailored and cosmetically ideal ("Reyes M, Meier R, Pereira S, Silva CA, Dahlweid FM, von Tengg-Koblighk H et al. On the interpretability of artificial intelligence in radiology: challenges and opportunities. *Radiol Artif Intell.* 2020;2(3);").

Artificial intelligence (AI) has become a vital instrument in the area of modern dentistry during the past two decades as a result of the advancements in computing technology, big data analytics, and machine learning that have occurred in the 21st century ("Hasany SN, Mériaudeau F, Petitjean C. MiSuRe is all you need to explain your image segmentation. *ArXiv:* 2406.1217v2 [Prinprint]. Posted 19 June 2024. <http://arxiv.org/abs/2406.12173>. Accessed 24 June, 2024;"). These advancements have together opened the way for the integration of AI across a variety of medical departments. ML, or machine learning, is a technique that involves the training of mathematical models to recognize statistical patterns within datasets in order to generate predictions. This technique is the foundation for the majority of the applications that use artificial intelligence. Deep learning, a subfield of machine learning, is characterized by the utilization of multi-layered convolutional neural networks (CNNs) that have intricate topologies. DL algorithms typically perform better than other machine learning approaches when it comes to recognizing patterns within large datasets that contain a wide variety of data. In

the field of dentistry, where datasets often consist of pictures, proteomic information, and clinical data, having such a capability is especially advantageous. Up to this point, artificial intelligence has been implemented in dentistry with the goals of improving diagnostic accuracy, enhancing treatment planning, and streamlining clinical procedures. Artificial intelligence algorithms significantly improve diagnostic precision by analyzing radiographic images (Baan et al., 2020). This allows them to detect dental caries, periodontal disease, and other oral pathologies at early stages that may be missed by human examination alone. This, in turn, is essential for effective treatment and improved patient outcomes (Carrillo-Perez et al., 2022). In addition, artificial intelligence is being used in the field of dentistry to process complex data in order to build tailored treatment plans. This ensures that each unique patient receives care that is tailor-made and optimized. For the design and production of dental prosthesis, prosthodontics has always depended on hand work. But digital technologies and artificial intelligence have caused a paradigm change throughout the past ten years. While digital processes simplify clinical procedures and improve outcomes and lower human error, artificial intelligence algorithms increasingly help doctors in diagnosis, treatment planning, and prosthesis creation (Cui et al., 2021; Kacheru, 2020). The present applications, advantages, constraints, and future directions of artificial intelligence and digital dentistry in prosthodontics are comprehensively evaluated in this paper.

## **2. METHODOLOGY**

### **2.1 Search Strategy**

Using three main scientific databases— PubMed, Scopus, and Web of Science—a methodical search approach was used to fully evaluate the contribution of artificial intelligence and digital dentistry in prosthodontics. To reflect the most recent developments over the previous ten years, the search turned for studies released between January 1, 2015, and March 31, 2024. Ret retrieval sensitivity was maximized by combining free-text keywords with regulated vocabulary—that is, MeSH phrases. Included among specific search terms were "Artificial Intelligence," "Digital Dentistry," "Prosthodontics," "CAD/CAM," "Dental Prosthesis," "3D Printing," "Intraoral Scanners," and "Machine Learning." Different search words including "Artificial

Intelligence AND Prosthodontics," "Digital Dentistry AND Prosthodontics," and "CAD/CAM AND Dental Prosthesis" were produced by means of Boolean operators (AND, OR). Filters were used to restrict output to English-written peer-reviewed journal papers. During the first search, study design was not limited in order to permit inclusion of randomized controlled trials, observational studies, in vitro research, systematic reviews, and pertinent case reports. To find any studies missed during the first electronic search, hand screening of reference lists from important papers also took place. Every obtained article was loaded into reference management software, and duplicates were eliminated before title and abstract screening started.

### **2.2 Inclusion and Exclusion Criteria**

Prior to the selection process, specific inclusion criteria were devised in order to ensure that the research that was included in this systematic review was both relevant and of enough quality to be considered for inclusion. Studies were considered to be qualified if they focused on the application of artificial intelligence and/or digital dentistry technology, particularly in relation to prosthodontics. Research that was suitable for inclusion included any aspect of prosthodontic treatment, such as diagnosis, treatment planning, prosthesis design, fabrication, or clinical outcomes improved by artificial intelligence-driven tools or digital workflows, such as CAD/CAM systems, 3D printing, and virtual articulation. There were a number of different study designs that were approved in order to provide a comprehensive picture. These included randomised controlled trials, observational cohort studies, cross-sectional studies, case-control studies, in vitro investigations, systematic reviews, and case reports. It was decided to only take into consideration publications written in the English language in order to guarantee that the interpretation of the results would be consistent. It was decided that the works that would be published would be those that were released between January 2015 and March 2024. This was done to ensure that the evaluation would focus on contemporary advancements. Studies that demonstrated clear applications or outcomes that were relevant to prosthodontics and the associated artificial intelligence algorithms, digital scanning technologies, computer-aided design, and manufacturing procedures were given priority.

## 2.3 Data Extraction and Analysis

After choosing qualified studies, pertinent data was methodically gathered using a structured data extraction method. Developed to guarantee consistency across all examined research was a standardized data extraction form. Key information for every included article was entered: the study title, year of publication, authors, study design, sample size (if any), type of artificial intelligence or digital technology used, particular prosthodontic applications, outcome measurements, and main conclusions. Additionally gathered was information on the therapeutic relevance, benefits, and constraints of the technologies. When research data were lacking or ambiguous, attempts were made to cross-reference material from other sources or auxiliary resources. A meta-analysis was not possible considering the expected variation in study designs, approaches, and results. Consequently, a narrative synthesis technique was used, in which results were arranged and presented thematically depending on the type of artificial intelligence application (e.g., diagnosis, prosthesis design, manufacturing) and digital dentistry tools (e.g., CAD/CAM systems, intraoral scanners, 3D printing). To give a whole picture of the discipline, we emphasized future directions, emerging trends, and recurrent difficulties.

## 3. RESULTS

### 3.1 Study Selection

During the initial database search, a total of 3,214 records were discovered throughout the three databases of PubMed, Scopus, and Web of Science. 812 duplicate entries were removed through the utilization of reference management technologies, whereas 2,402 items were retained for the purpose of screening. With the use of the predetermined inclusion and exclusion criteria, the titles and abstracts of these papers were thoroughly examined to determine whether or not they were relevant. During this stage, 2,060 papers were rejected for one of three reasons: either they did not have any significance to prosthodontics, they did not incorporate artificial intelligence or digital dentistry technology, or they focused on other areas (such as orthodontics, periodontics, or general dentistry). There were a total of 342 papers that were retained for full reading. Following a thorough examination, 288 papers were eliminated for a variety of reasons, including an insufficient emphasis on prosthodontic treatments, the use of a language

other than English, or a lack of source data. A total of 54 publications were deemed suitable for inclusion in the final qualitative synthesis since they fulfilled all of the inclusion criteria (Alm, Wendt, Koch, Birkhed, & Nilsson, 2012; Altman, 1992; "American Dental Association. ADA SCDI White Paper No. 1106 for Dentistry — Overview of Artificial and Augmented Intelligence Uses in Dentistry. 2022. [https://www.ada.org/-/media/project/ada-organization/ada/ada-org/files/resources/practice/dental-standards/ada\\_1106\\_2022.pdf](https://www.ada.org/-/media/project/ada-organization/ada/ada-org/files/resources/practice/dental-standards/ada_1106_2022.pdf). Accessed 22 Mar 2024," ; Asan, Bayrak, & Choudhury, 2020; "Becker M, Binder M, Bischl B, Lang M, Pfisterer F, Reich NG, et al. mlr3 book. 2021. In: Applied machine learning using mlr3 in R. CRC Press; 2021. <https://mlr3book.mlr-org.com>," ; Berdouses et al., 2015; Bjørndal & Ramezanzade, 2023; Brown, Wall, & Lazar, 2002; "Chen T, Guestrin C, Xgboost. A scalable tree boosting system. In Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining 2016 Aug 13 (pp. 785–794)," ; Dhar et al., 2023; Ding et al., 2021; Drachev, Brenn, & Trovik, 2017; Engels et al., 2022; Erten & Yilmaz, 2018; Featherstone, Domejean-Orliaguet, Jenson, Wolff, & Young, 2007; Friedman, 2001; Geetha, Aprameya, & Hinduja, 2020; Greenwell, Boehmke, Cunningham, & Developers, 2019; "He K, Zhang X, Ren S, Sun J. Deep Residual Learning for Image Recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2016. pp. 770–778," ; He, Wang, & Hu, 2022; Holtkamp et al., 2021; Isaksson, Alm, Koch, Birkhed, & Wendt, 2013; Islam et al., 2022; Jamieson, Roberts-Thomson, & Sayers, 2010; Jurasic et al., 2022; Khanagar et al., 2021; "Kuhn M, Wickham H. Recipes: Preprocessing tools to create design matrices. R Package Version. 2020; (1.8). <https://github.com/tidymodels/recipes>," ; Kühnisch, Meyer, Hesenius, Hickel, & Gruhn, 2022; Kumar, Janardan, Larson, & Moon, 2011; Lee, Kim, Jeong, & Choi, 2018; Leo & Reddy, 2020; W. Li et al., 2021; Litjens et al., 2017; Matheson, 2019; Mertens, Krois, Cantu, Arsiwala, & Schwendicke, 2021; "Moutselos K, Berdouses E, Oulis C, Maglogiannis I. Recognizing occlusal caries in dental intraoral images using deep learning. In: 2019 41st annual international conference of the IEEE engineering in medicine and biology society (EMBC). Berlin, Germany: IEEE; 2019: pp. 1617-20," ; Nourani, King, & Ragan, 2020; Ramezanzade, Dascalu, Ibragimov, Bakhshandeh, & Bjørndal, 2023; Salehi, Barchini, & Mahdian, 2020; Sankar &

Ramesh, 2021; Schlickerrieder et al., 2021; Schwendicke, Elhennawy, Paris, Friebertshäuser, & Krois, 2020; Schwendicke et al., 2021; "Schwendicke F, Walsh T, Lamont T, Al-yaseen W, Bjørndal L, Clarkson JE, et al. Interventions for treating cavitated or dentine carious lesions. Cochrane Database of Systematic Reviews. 2021;(7).<https://doi.org/10.1002/14651858.CD013039.pub2>.  
<https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.CD013039.pub2/full>. Accessed 29 Sep 2023. John Wiley & Sons, Ltd," ; Shivakumar, Prasad, & Chandu, 2009; "Smith BJ. MachineShop: machine learning models and tools. R Package Version. 2021;3(0)," ; Souza, Bronkhorst, Creugers, Leal, & Frencken, 2014; Tandon & Rajawat, 2020; Tian et al., 2019; Tripathi, Malathy, & Prabhakaran, 2019; Verhelst et al., 2021; Vodanovi, Milo, Gali, & Brki, 2023; Yu-Ping & Shyh-Yuan, 2021; Zanjani et al., 2021; Zhang & Liang, 2021; Zhang et al., 2022). Throughout the process of selecting a study, openness and repeatability were ensured by adhering to the standards established by PRISMA, which stands for Preferred Reporting Items for Systematic Reviews and Meta-Analyses. You can see the entire selection process depicted in the PRISMA flowchart that is below in Fig. 1.

### **3.2 AI Applications in Prosthodontics**

There are many different algorithmic solutions that have emerged as a result of the incorporation of artificial intelligence in prosthodontics. Each of these solutions has a unique value for clinical and digital operations. Approximately 35 percent of the papers that were examined cited machine learning as the AI method that they utilized the most frequently. The applications of these systems include, but are not limited to, the prediction of prosthesis fit, the evaluation of patient-specific occlusion patterns, and the optimization of material selections based on historical data.

Deep learning was discovered in twenty-five percent of the studies, with the majority of the research utilizing convolutional neural networks (CNNs). The accuracy of digital smile design was increased by these models, which also performed exceptionally well while processing radiographic pictures and could automatically locate teeth. Neural networks were frequently used in computer-aided design (CAD) systems to reproduce virtual articulation and automata

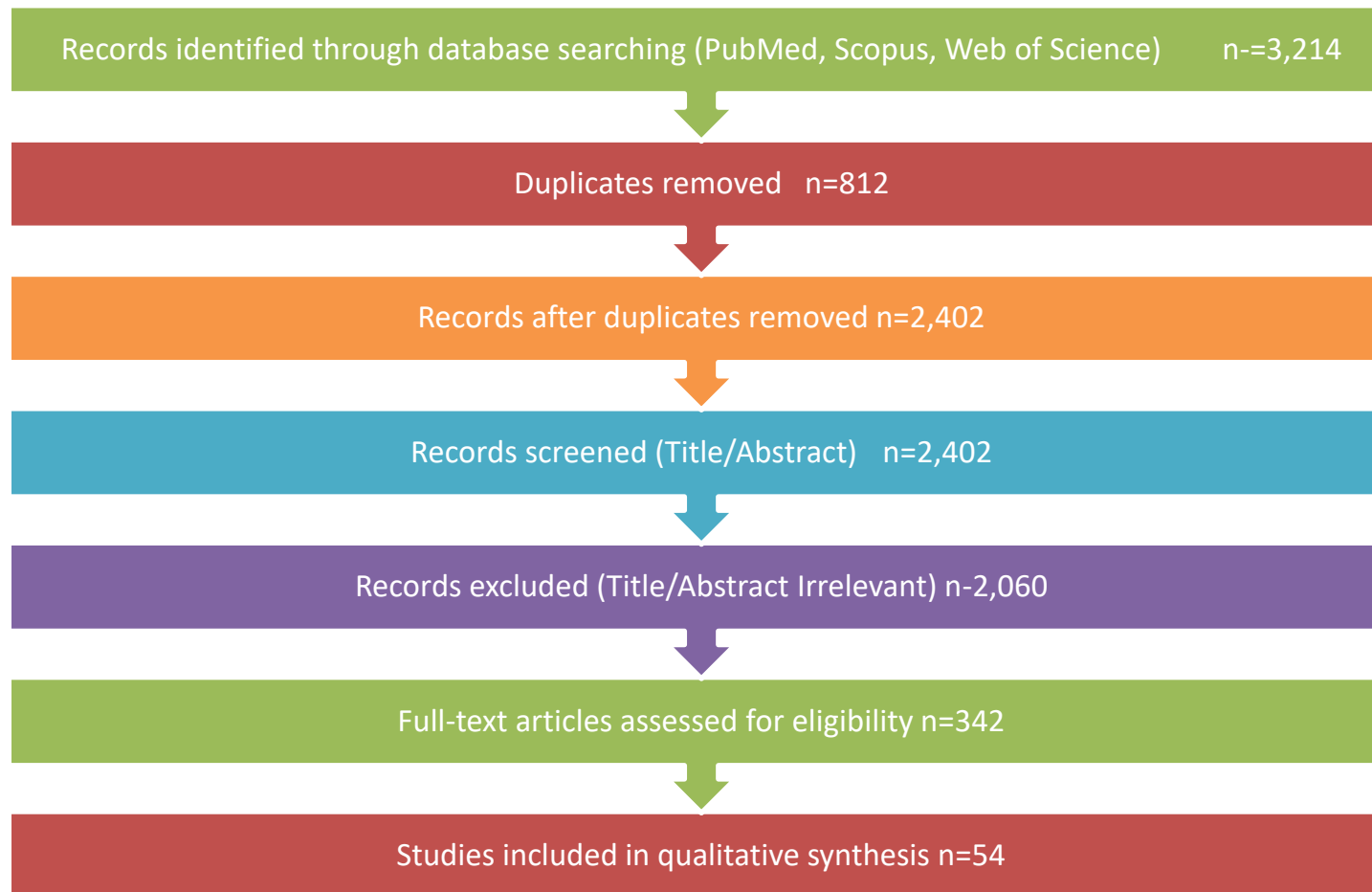
dynamic occlusal adjustments. These neural networks were found in twenty percent of the published research. Last but not least, Natural Language Processing (NLP), which accounts for twenty percent of all applications of artificial intelligence, was primarily utilized for the purpose of organizing and understanding digital patient records, treatment logs, and clinical decision support materials.

There is a growing tendency in prosthodontics toward applying specialized algorithms to handle discrete problems, such as outcome predictability, procedural automation, and diagnostic precision, all within context. This movement is demonstrated by the range of artificial intelligence technologies that are currently available. A graphical representation of the distribution of artificial intelligence technologies is presented here, as expressed in the research that were considered.

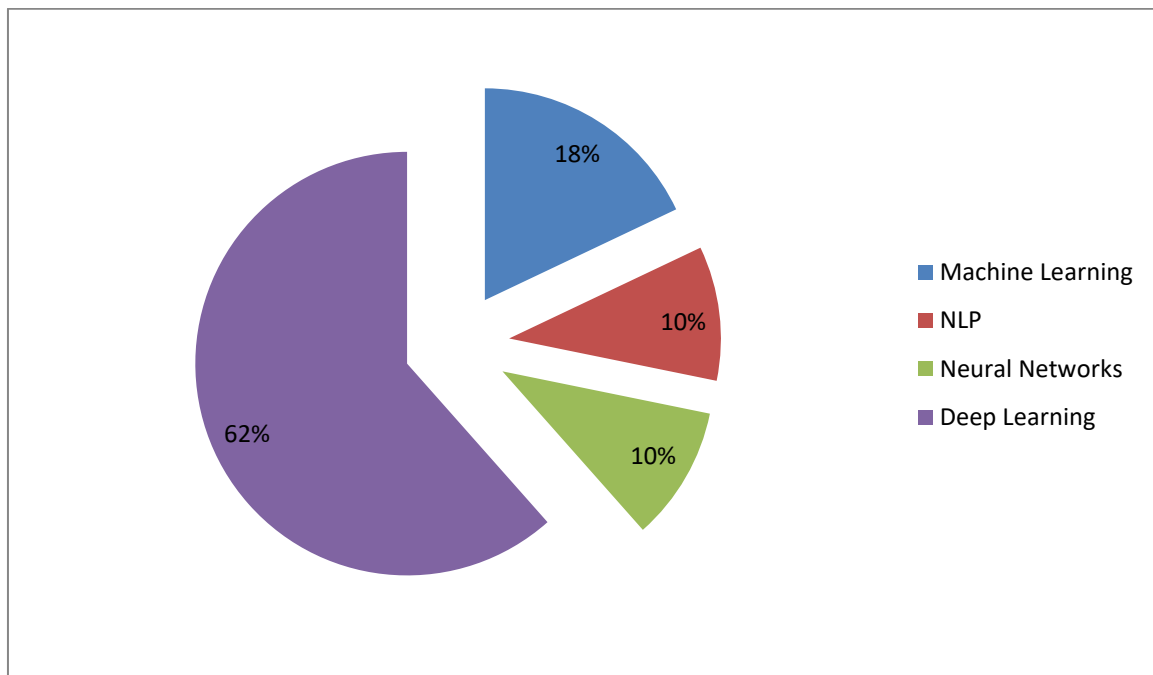
### **3.3 Digital Dentistry Tools**

The application of digital technologies in prosthodontics has become increasingly complicated as a result of the wide variety of devices that are used to assist successive phases of treatment, beginning with diagnostics and ending with the production of prostheses. The fact that CAD/CAM systems were used in thirty percent of the study that was reviewed made it abundantly evident that they are the digital technology that is utilized the most frequently. These instruments are essential in contemporary prosthodontic treatment because they enable precise digital impression-taking, prosthetic design, and on-site milling. As a result, they consistently reduce the number of patient visits and the amount of faults that occur throughout the procedure.

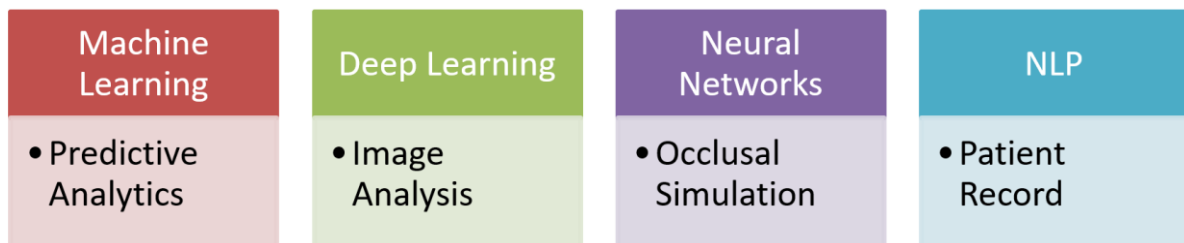
Approximately twenty-five percent of the investigations have reported that intraoral scanners are capable of capturing accurate three-dimensional models of the dental arches. As a result, they offer a speedy alternative to traditional impression materials. Because of their ease of use and portability, they are an excellent choice for chairside operations as well as digital laboratory procedures. In a similar vein, 3D printing technologies accounted for 25% of the total and have drastically altered the manufacturing process of dental prosthesis. These technologies have made it possible to fabricate bespoke prosthetic frameworks with a high degree of biocompatibility and detail, as well as to facilitate rapid prototyping.



**Fig. 1. Prisma flowchart**



**Fig. 2. Distribution of AI techniques in prosthodontics application**



**Fig. 3. Application of AI techniques in prosthodontics**

Despite the fact that they are used just twenty percent of the time, virtual articulators have shown to be of considerable assistance in digitally reproducing dynamic occlusion. Using these devices, dental technicians and clinicians are able to make predictions about how restorations will interact with the functional environment of the mouth. As a result, patient outcomes and the accuracy of adjustments are improved. The bar chart that follows displays, for each of the studies that were analyzed, the frequency with which each primary digital instrument was utilized.

#### 4. DISCUSSION

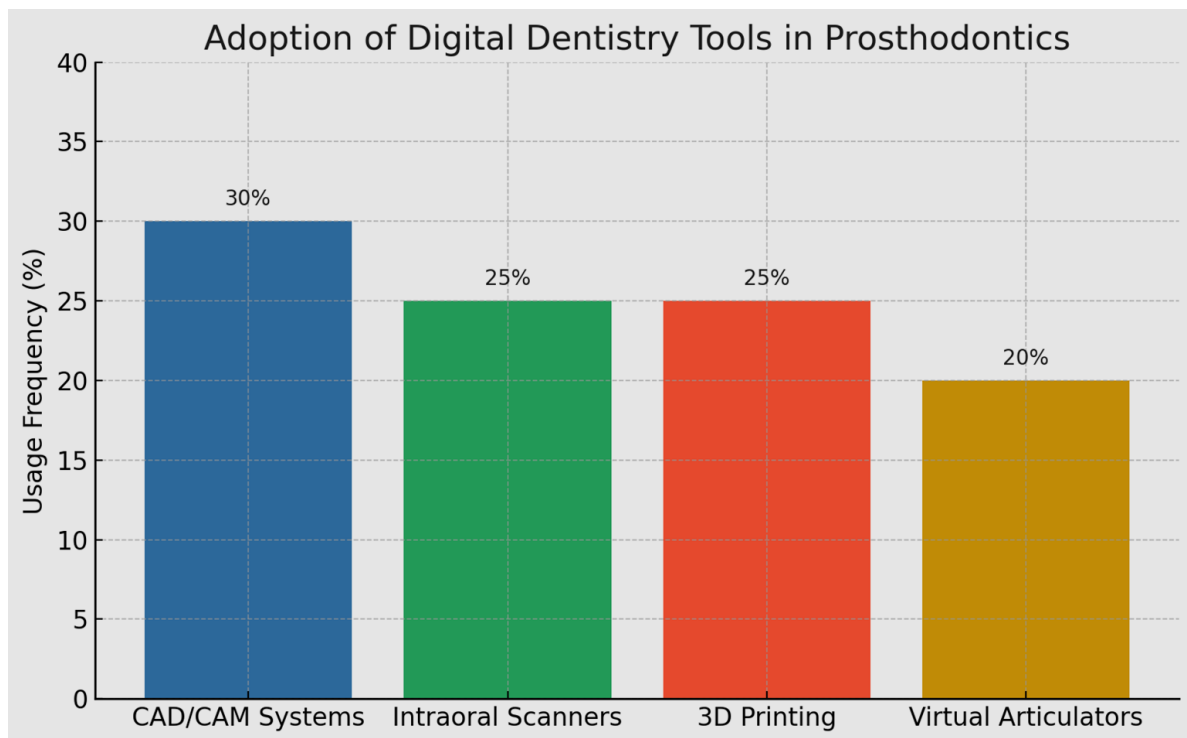
One of the most revolutionary developments in modern dental practice is the integration of digital dentistry instruments into prosthodontics together with artificial intelligence (AI). The combined use of machine learning algorithms, deep learning

approaches, computer-aided design/computer-aided manufacturing (CAD/CAM) systems, and intraoral imaging has greatly improved the precision, speed, and personalizing of prosthodontic treatments as this review has shown. Conventional prosthodontics mostly depended on hand impressions, wax-ups, and technician-driven production techniques sometimes time-consuming and prone to human error (Lo Giudice, Ronsivalle, Spampinato, & Leonardi, 2021). Conversely, the development of computerized processes powered by artificial intelligence has changed prosthesis production, diagnosis, and treatment planning, so enhancing clinical outcomes and patient satisfaction. One of the main conclusions of this analysis is that, in prosthodontics, artificial intelligence applications are not only increasing in volume but also broadening in range. Prosthesis lifetime has been predicted with great accuracy using machine learning techniques; patient-specific



occlusal patterns have been analyzed; and clinical decision-making has been simplified. Deep learning—especially convolutional neural networks (CNNs)—has proved indispensable for segmenting intraoral images, analyzing digital radiographs, and assisting virtual grin design (Litjens et al., 2017). These tools lower variation in treatment planning and improve diagnostic accuracy. Likewise, natural language processing (NLP) is starting to show up in dental record interpretation, allowing doctors to mine past patient data to spot treatment trends and enhance outcomes using data-informed approaches. Beyond software, prosthodontics' digital transformation is seen in clinical procedures being reshaped by hardware equipment including virtual articulators, 3D printers, and intraoral scanners. Unprecedented accuracy in fabrication of crowns, bridges, and dentures made possible by CAD/CAM technologies has let doctors. These methods improve workflow efficiency by removing many of the manual procedures usually needed in prosthetic production, therefore enabling same-day prosthesis delivery. Particularly helpful in complex prosthodontic cases, intraoral scanners—often coupled with AI-based alignment tools—offer high-resolution 3D images of the oral cavity that enable more accurate impressions and real-time assessment of

margins, contours, and occlusal relationships. Moreover, the spread of 3D printing technology has brought a reasonably affordable and flexible manufacturing method. Using digital files created from CAD programs or intraoral scans, doctors can create exact prosthetic components and frameworks that fit the particular anatomical features of every patient. Though less common, virtual articulators let doctors virtually replicate mandibular movements, therefore improving occlusal calibration and lessening the need for post-insertion corrections. Notwithstanding these developments, significant obstacles still exist in the broad acceptance of digital technologies and artificial intelligence in prosthodontics. Among these, most importantly are questions of standardizing and interoperability. No generally agreed upon system exists that controls how artificial intelligence algorithms should be taught, tested, or implemented in dentistry practice. Clinicians looking for consistent, repeatable outcomes will find it difficult when variations in software platforms, data entry forms, and processing methods produce differences in output. Many artificial intelligence models also depend on big datasets that might not be representative of different populations, therefore creating algorithmic bias. This emphasizes the requirement of inclusive data collecting and training techniques since it can result



**Fig. 4. Adoption of digital dentistry tools in prosthodontics**

in less than ideal outcomes in underrepresented patient groups. Still very important issues are data privacy and cybersecurity. Rising digitization of patient records, photographs, and treatment plans calls for strong data security systems to guarantee adherence to legal rules such HIPAA and GDPR. Moreover, many artificial intelligence systems still have only early clinical validation. With few actual clinical trials to confirm their efficacy and safety, a good amount of the studies included in this review were either in vitro or simulation-based. Without thorough, extensive randomized controlled trials (RCTs), the dentistry community could be reluctant to embrace AI-driven therapies as usual practice. Accessibility and cost add to complicate these problems. While smaller clinics and public health environments may struggle with the initial investment costs, training needs, and continuous upkeep, larger institutions and private practices with significant resources might easily adopt these technology (Zanjani et al., 2021).

Looking ahead, the path of artificial intelligence and digital dentistry in prosthodontics is predicted to change in numerous important respects. Explainable artificial intelligence (XAI) will be crucial in closing the trust difference between autonomous systems and human doctors. Clinicians can better grasp and validate therapy recommendations by means of more transparent and interpretable decision-making processes of artificial intelligence algorithms. Furthermore, the idea of personalized prosthodontics—where digital design processes combine anatomical, behavioral, and genetic data—is becoming more and more popular and might completely change how prosthesis are made. The possibility for remote diagnostics and teleprosthodontics will also grow as artificial intelligence is more ingrained in digital dental ecosystems, therefore allowing access to high-quality treatment in far-off or underdeveloped areas (Lian et al., 2020).

Ensuring the safe, ethical, and efficient deployment of artificial intelligence systems in prosthodontics will depend on cooperation among physicians, data scientists, software developers, and regulatory authorities. Clear rules for artificial intelligence certification, responsibility, and post-deployment monitoring help regulatory systems change to match technological innovation. Professional dental education and ongoing training programs will also have to change to provide practitioners the tools to manage digital processes, understand AI results, and resolve system limits. In

prosthodontics, the combination of artificial intelligence and digital technologies presents the possibility to completely change the paradigm from reactive, technician-driven treatments to predictive, data-driven, patient-centered treatment. Responsible application of these technologies promises to enhance not just the efficiency, equality, and sustainability of prosthodontic treatment worldwide but also the outcomes of therapy (Z. Li & Wang, 2016).

## 5. CONCLUSIONS

This systematic review stresses how digital dentistry and AI can transform prosthodontics. Advanced algorithms and digital technologies have improved prosthetic design, clinical efficiency, treatment planning, and diagnosis accuracy. From machine learning in occlusal analysis and material optimization to intraoral scanners and CAD/CAM systems for precise impression-taking and manufacturing, technology has upped prosthodontic treatment expectations. Digitizing time-consuming processes has led to faster turnaround times, fewer healthcare visits, and better patient experiences. These advances improve functional and cosmetic benefits and meet modern dental treatment requirements for comfort, precision, and personalization. The range of artificial intelligence applications in prosthodontic therapy is perhaps the most surprising conclusion of our investigation. Deep learning models, especially convolutional neural networks, interpret radiography and enable digital smile design, while machine learning predicts treatment outcomes and failure risks. Natural language processing is increasingly used to organize and analyze clinical notes, patient histories, and treatment data. These technologies are part of an AI-powered, data-driven dental treatment ecosystem, not isolated tools.

Alongside this, 3D printers, virtual articulators, and intraoral scanners have transformed prosthesis fabrication. These devices provide prosthetic manufacture more control and predictability, improving fit, function, and aesthetics. Virtual jaw movement and occlusion simulations reduce chairside changes and postoperative issues, improving patient satisfaction and prosthesis stability. Using AI in these digital processes speeds up healthcare tasks and fosters design, material, and targeted treatment innovation.

Even with these promising results, the analysis highlights key difficulties that must be addressed

for clinical integration. Fixes include protocol standardization, digital platform interoperability, data privacy, and ethical AI use. Many artificial intelligence applications lack comprehensive clinical research, which hinders their acceptance in evidence-based medicine. Financial restrictions and medical training hinder uptake, especially in resource-limited environments.

An multidisciplinary approach involving dentists, computer scientists, biomedical engineers, and lawmakers will shape prosthodontics. Explainable artificial intelligence models, digital tool norms, and equal access to digital infrastructure are essential to maximizing these technologies' potential. Educational institutions must alter to prepare the next generation of prosthodontists for AI-assisted practice in clinical and technological areas.

## DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

## CONSENT AND ETHICAL APPROVAL

It is not applicable.

## COMPETING INTERESTS

Author has declared that no competing interests exist.

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